#OHBM2019
@JoramSoch

Abstract #3967
Poster #M495

The problem of correlated trial-by-trial parameter estimates in fMRI decoding – solved

Joram Soch[1,2,•], Carsten Allefeld[1,2], John-Dylan Haynes[1–8]

Bernstein Center for Computational Neuroscience Berlin

[1] Bernstein Center for Computational Neuroscience, [2] Berlin Center for Advanced Neuroimaging, [3] Berlin School of Mind and Brain, Berlin, Germany
[4] Clinic for Neurology, Charité – Universitätsmedizin Berlin, Germany, [5] Department of Psychology, Humboldt-Universität zu Berlin, Germany
[6] EXC NeuroCure, Charité Berlin, [7] EXC Science of Intelligence, TU Berlin, [8] CRC Volition and Cognitive Control, TU Dresden, Germany
• BCCN Berlin, Philippstraße 13, Haus 6, 10115 Berlin, Germany / **joram.soch@bccn-berlin.de**

CHARITÉ UNIVERSITÄTSMEDIZIN BERLIN

Introduction

Techniques of multivariate pattern analysis (MVPA) can be used to decode the discrete experimental condition or a continuous modulator variable from measured brain activity during a particular trial. In functional magnetic resonance imaging (fMRI), trial-wise response amplitudes are sometimes estimated from the BOLD signal using a general linear model (GLM) with one onset regressor for each trial [1]. When using rapid event-related designs with trials closely spaced in time, those estimates are highly variable and serially correlated due to the delayed shape of the hemodynamic response function (HRF) [2,3,4]. Here, we describe inverse transformed encoding models (ITEM) [10], a principled approach of accounting for those serial correlations and decoding from the resulting estimates.

Theory

In univariate fMRI data analysis, the standard general linear model is commonly used to estimate condition-wise activation patterns $\hat{\beta}$ on which statistical inference is then based:

$$y = X\beta + \varepsilon,\ \varepsilon \sim \mathrm{N}(0, \sigma^2 V)$$
$$\hat{\beta} = (X^T V^{-1} X)^{-1} X^T V^{-1} y$$

In multivariate fMRI data analysis, a trial-wise general linear model is sometimes used to obtain trial-wise response amplitudes $\hat{\gamma}$ on which decoding analyses are later performed:

$$y = X_t\gamma + \varepsilon_t,\ \varepsilon_t \sim \mathrm{N}(0, \sigma_t^2 V)$$
$$\hat{\gamma} = (X_t^T V^{-1} X_t)^{-1} X_t^T V^{-1} y$$

We define a transformation matrix T that maps from the $n \times t$ trial-wise design matrix X_t to the $n \times p$ design matrix X (see Figure 1):

$$X = X_t\, T$$

We also define the uncorrelation matrix U as the $t \times t$ covariance matrix of the trial-wise parameter estimates $\hat{\gamma}$ (see Figure 2):

$$U = (X_t^T V^{-1} X_t)^{-1}$$

The transformation matrix T and the uncorrelation matrix U can be used to derive a new GLM, the transformed encoding model that directly operates on the trial-wise response amplitudes $\hat{\gamma}$:

$$\hat{\gamma} = T\beta + \eta,\ \eta \sim \mathrm{N}(0, \sigma^2 U)$$
$$\hat{\beta} = (T^T U^{-1} T)^{-1} T^T U^{-1} \hat{\gamma}$$

When obtaining trial-wise parameter estimates $\hat{\Gamma} = [\hat{\gamma}_1, \ldots, \hat{\gamma}_v]$ from multiple voxels, e.g. a searchlight or region of interest, this leads to a multivariate transformed encoding model (see Figure 3A):

$$\hat{\Gamma} = TB + H,\ H \sim \mathrm{MN}(0, U, \Sigma_y)$$

Because our goal here is not statistical inference, i.e. describing the trial-wise response amplitudes $\hat{\Gamma}$ in terms of the experimental design T, but decoding analysis, i.e. recovering the experimental design from trial-wise response amplitudes [5], we turn this model into an inverse transformed encoding model (see Figure 3B):

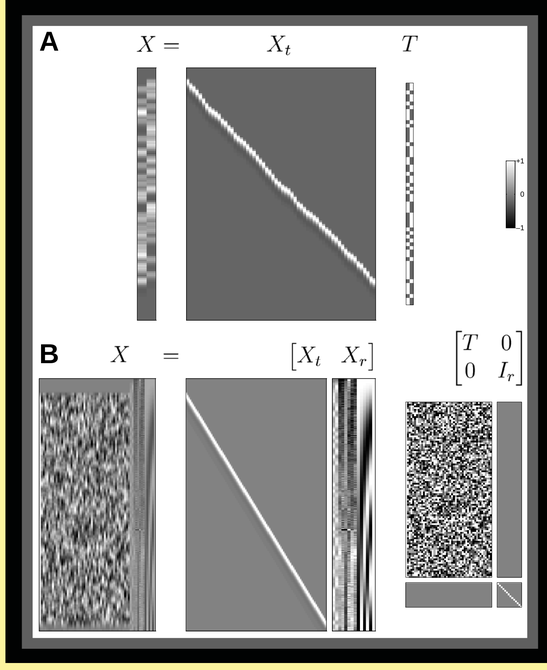
$$T = \hat{\Gamma} W + N,\ N \sim \mathrm{MN}(0, U, \Sigma_x)$$

To assess decoding accuracy, the ITEM is estimated by cross-validation across fMRI recording sessions. The weight matrix W is estimated from all but one session ($\neg j$) and design variables T are decoded by out-of-sample prediction in the left-out session (j):

$$\hat{W}_{\neg j} = \left(\hat{\Gamma}^T U^{-1} \hat{\Gamma}\right)^{-1} \hat{\Gamma}^T U^{-1} T$$
$$\hat{T}_j = \hat{\Gamma}_j\, \hat{W}_{\neg j}$$

From the reconstructed transformation matrix $\hat{T}$, one can read out trial-by-trial information (see Figure 3B), i.e. experimental condition or parametric modulator. The decoding accuracy is quantified as proportion correct for discrete classification between experimental conditions (see Simulation and Figure 4/5) or via correlation coefficients for continuous reconstruction of parametric modulators (see Application and Figure 7/8).

Methods



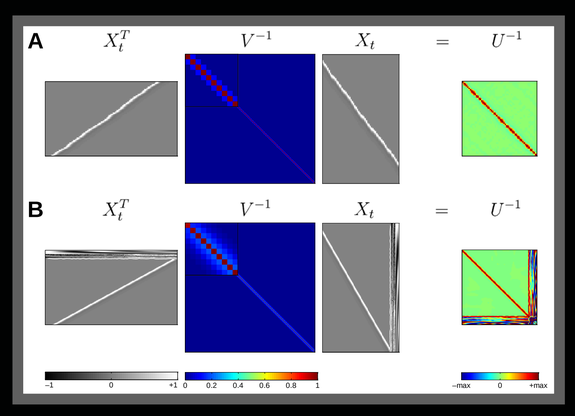
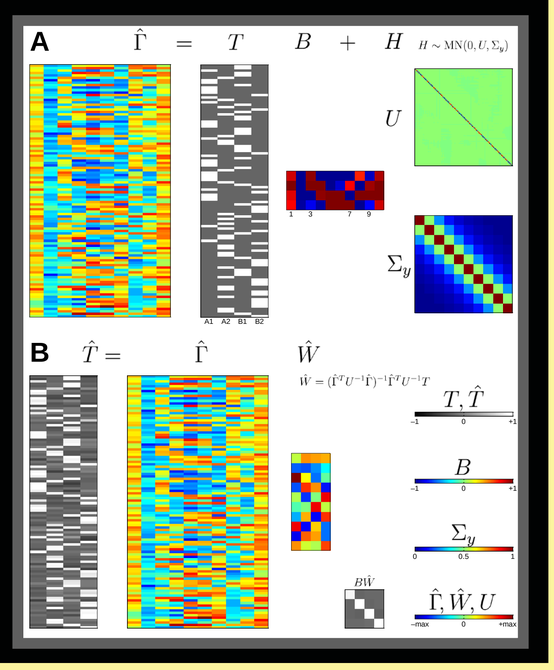


Figure 2 (↑). The uncorrelation matrix.

Figure 1 (←). The transformation matrix.

Figure 3 (→). A hypothetical reconstruction.



Simulation

In order to validate our method, we repeat a simulation described by Mumford et al. [2] and compare three approaches:

- the naïve approach ignoring trial-by-trial correlations („least squares, all"; LS-A);
- the standard approach using separate designs („least squares, separate"; LS-S);
- the proposed approach accounting for trial-by-trial correlations (ITEM) [10].

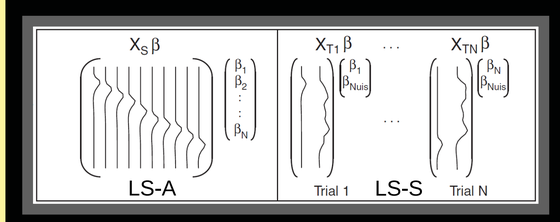


Figure. Trial-wise response estimation [2].

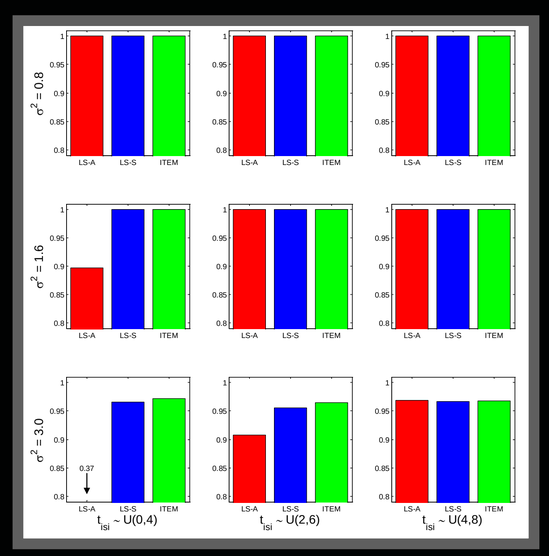


Figure 4. Simulation: statistical power.

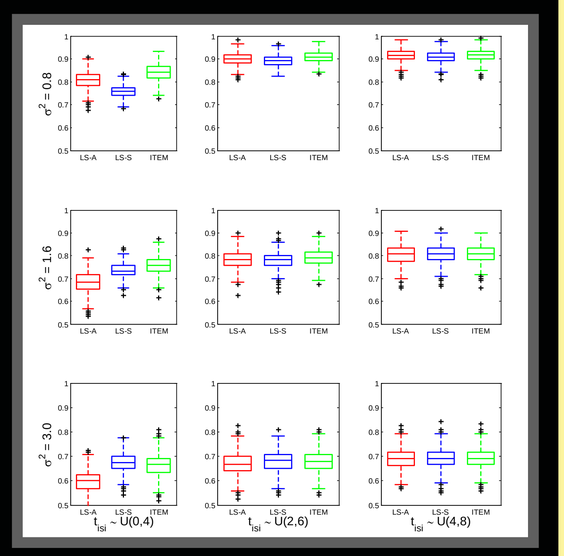


Figure 5. Simulation: decoding accuracies.

Application

In order to test our method, we re-analyze data on visual receptive fields acquired by Heinzle et al. [6]; fMRI data were converted to the BIDS format [7], preprocessed using SPM12 [8] and further analyzed using the MACS toolbox [9]. Then, ROI-based ITEM analysis was applied [10].

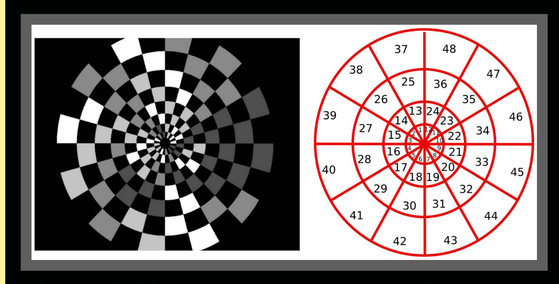


Figure 6. Experimental paradigm [6].

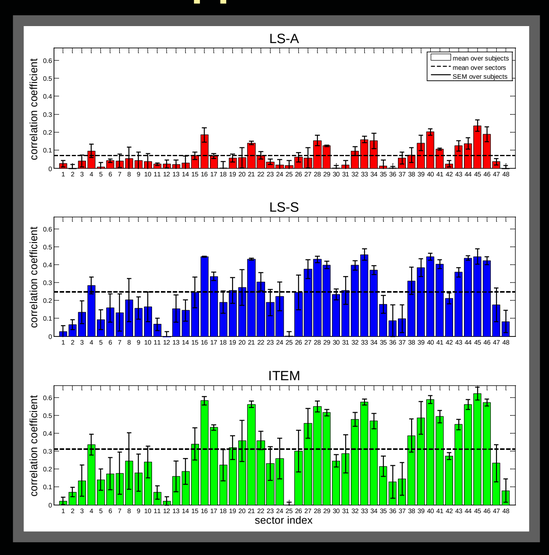


Figure 7. Application: decoding accuracies.

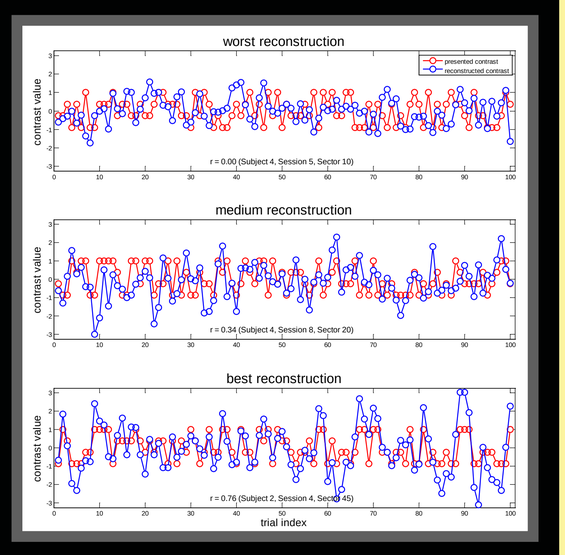


Fig. 8. Application: example reconstructions.

Download

(scan, if poster; click, if PDF)

Preprint @ bioRxiv

Toolbox @ GitHub

Discussion

The problem of correlated trial-by-trial parameter estimates has been discussed several times in the fMRI/MVPA literature [2,3,4]. Whereas the previous contributions have suggested ad-hoc solutions, e.g. estimating each trial using a separate design matrix [2], the present work provides a principled approach, based on the actual distribution of the trial-wise parameter estimates, as implied by the trial-wise design matrix [10]. However, ITEMs are not only applicable to rapid event-related designs, but generally useful when trial-wise linear decoding by classification or reconstruction is the goal. We hope that the ITEM approach – available as an SPM plug-in on GitHub – will increase the power of fMRI-based MVPA.

References

(1) Rissman J, Gazzaley A, D'Esposito M (2004). Measuring functional connectivity during distinct stages of a cognitive task. NeuroImage, vol. 23, pp. 752-763.
(2) Mumford JA, […], Poldrack RA (2012). Deconvolving BOLD activation in event-related designs for multivoxel pattern classification analyses. NeuroImage, vol. 59, pp. 2636-2643.
(3) Turner BO, […], Ashby FG (2012). Spatiotemporal activity estimation for multivoxel pattern analysis with rapid event-related designs. NeuroImage, vol. 62, pp. 1429-1438.
(4) Weeda W (2018). Estimating Single-trial BOLD Amplitude and Latency in Task-based fMRI Data with an Unknown HRF. OHBM 2018, Poster #2532.
(5) Haufe S, […], Bießmann F (2014). On the interpretation of weight vectors of linear models in multivariate neuroimaging. NeuroImage, vol. 87, pp. 96-110.
(6) Heinzle J, Kahnt T, Haynes JD (2011). Topographically specific functional connectivity between visual field maps in the human brain. NeuroImage, vol. 56, pp. 1426-1436.
(7) Gorgolewski KJ et al. (2016). The brain imaging data structure, a format for organizing and describing outputs of neuroimaging experiments. Scientific Data, vol. 3, art. 160044.
(8) Ashburner J et al. (2017). SPM12 Manual; URL: http://www.fil.ion.ucl.ac.uk/spm/doc/spm12_manual.pdf.
(9) Soch J, Allefeld C (2018). MACS – a new SPM toolbox for model assessment, comparison and selection. Journal of Neuroscience Methods, vol. 306, pp. 19-31.
(10) Soch J, Allefeld C, Haynes JD (in review). Inverse Transformed Encoding Models – a solution to the problem of correlated trial-by-trial parameter estimates in fMRI decoding; in review at NeuroImage, available from bioRxiv; DOI: 10.1101/610626; URL: https://www.biorxiv.org/content/10.1101/610626v1.